

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

INDEX

Sr. No.	Description	Page No.
1.	Methodology	1
	1 Satellite SAR System with Resolution and Revisit Time	
	2 Retrieving Multi-Temporal SAR Imagery Over the Specified Period	
	3 Differential SAR Interferometry (D-InSAR) for Displacement Measurement	2
2.	Visualization	3
	1 Defining specific bridge location & parameters for satellite SAR based inspection	
	2 Calibration, georeferencing, orthorectification of SAR images (VH/VV)	4
	3 Results of SAR based bridge inspection, including visualizations	5
	3.1 VV Displacement From 23 Jan 2025 to 28 June 2025	6
	3.2 VH Displacement From 23 Jan 2025 to 28 June 2025	6
	3.3 SAR Based Analysis at Abutment 1 LHS	7
	3.4 SAR Based Analysis at Abutment 1 RHS	8
	3.9 SAR Based Analysis at Abutment 2 LHS	9
	3.10 SAR Based Analysis at Abutment 2 RHS	10
3.	Analysis	11
	1 Identify potential deformation, shifts or movement of the bridge structure	
	Conclusion	

METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

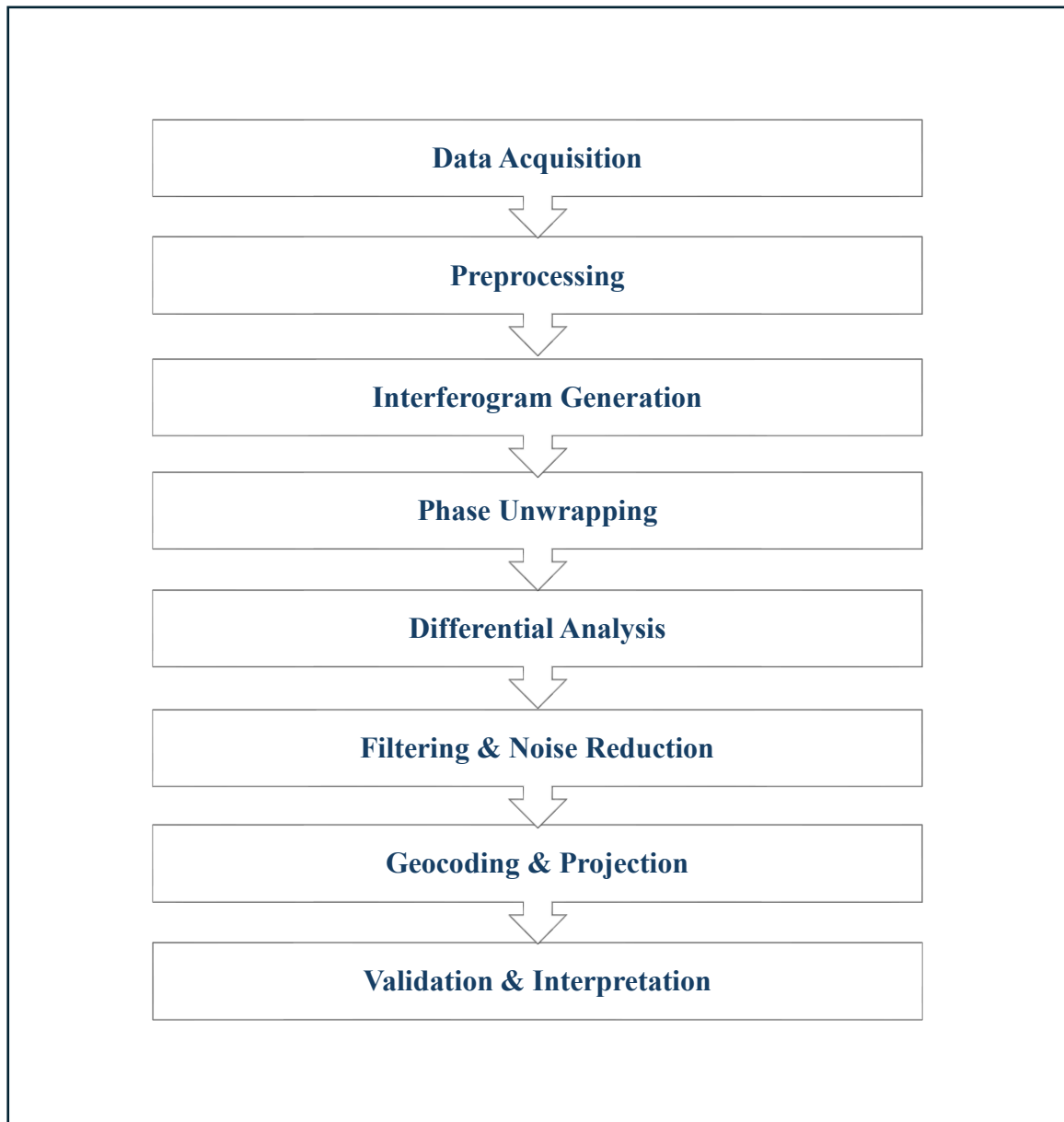
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 941+300

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

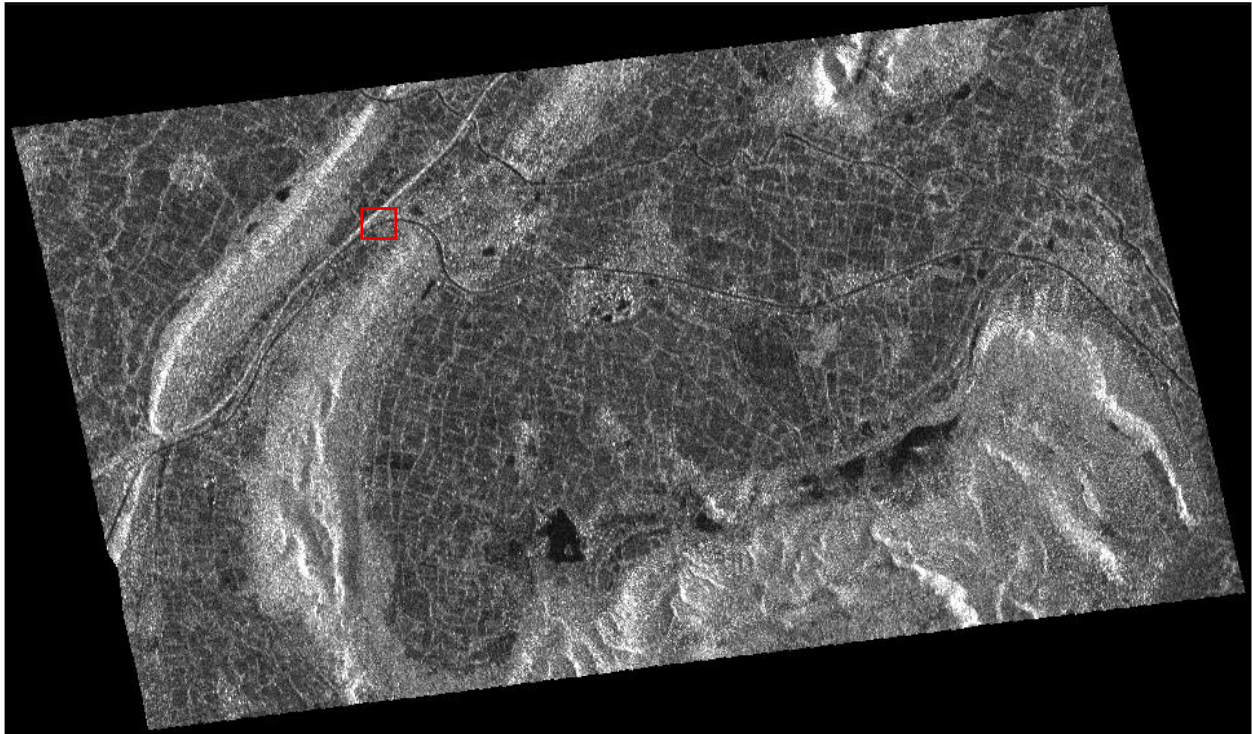
Flight Direction - DESCENDING

Polarization – VV+VH

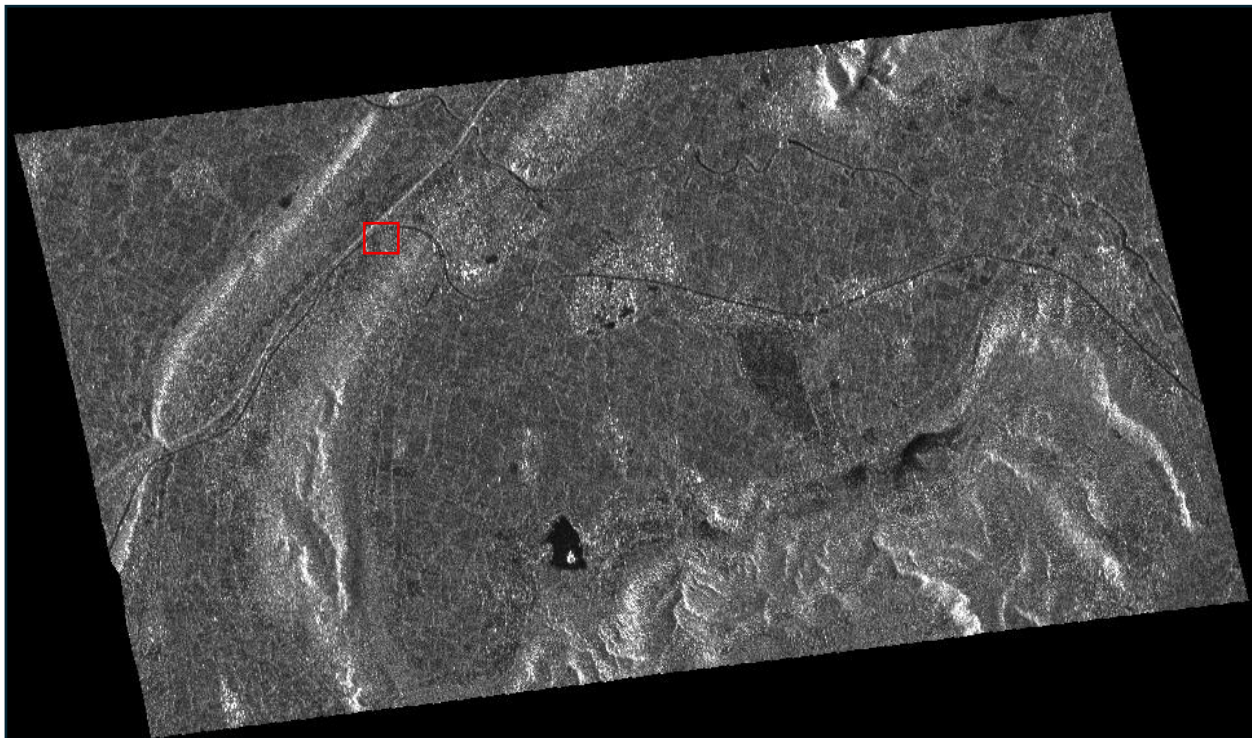
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

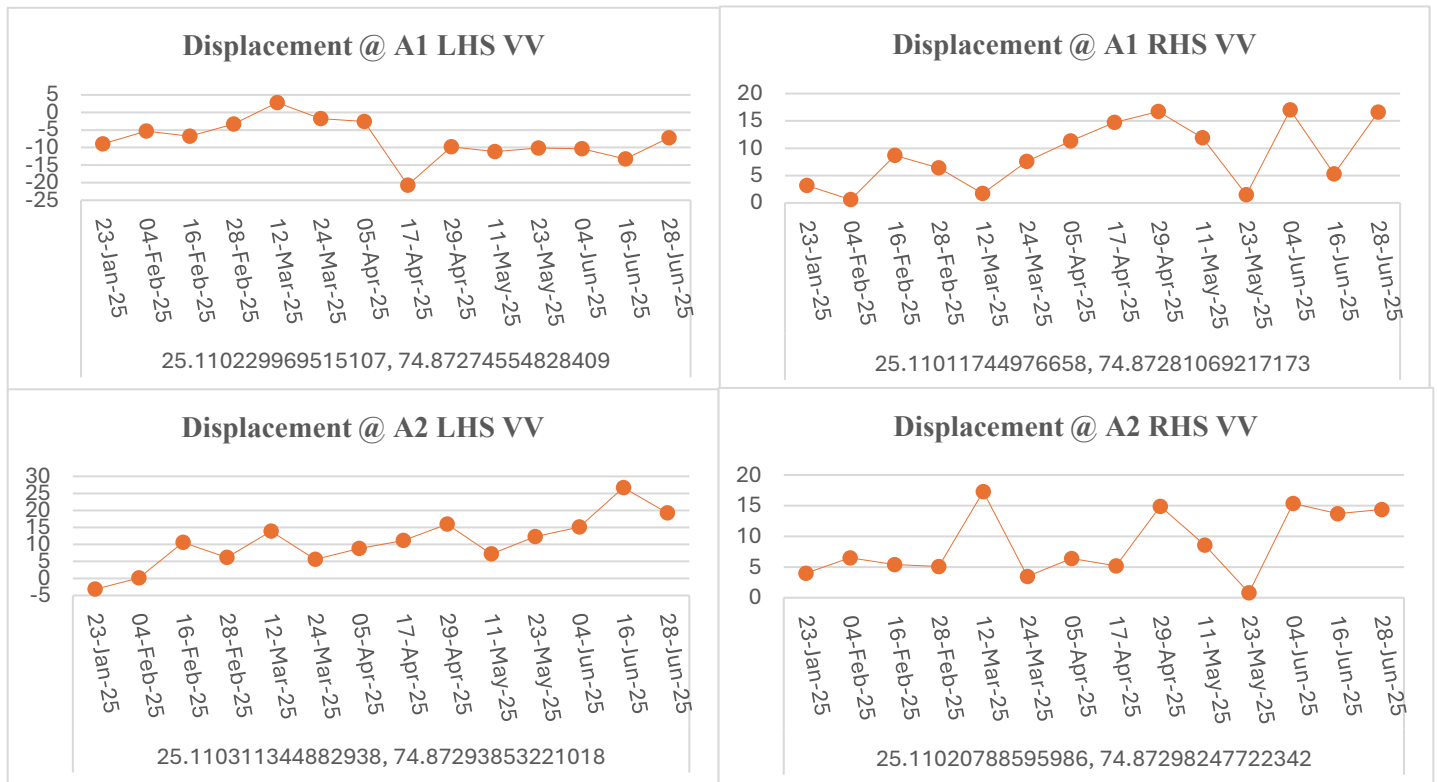
MNB – CH-941+300

Length – 32.69M

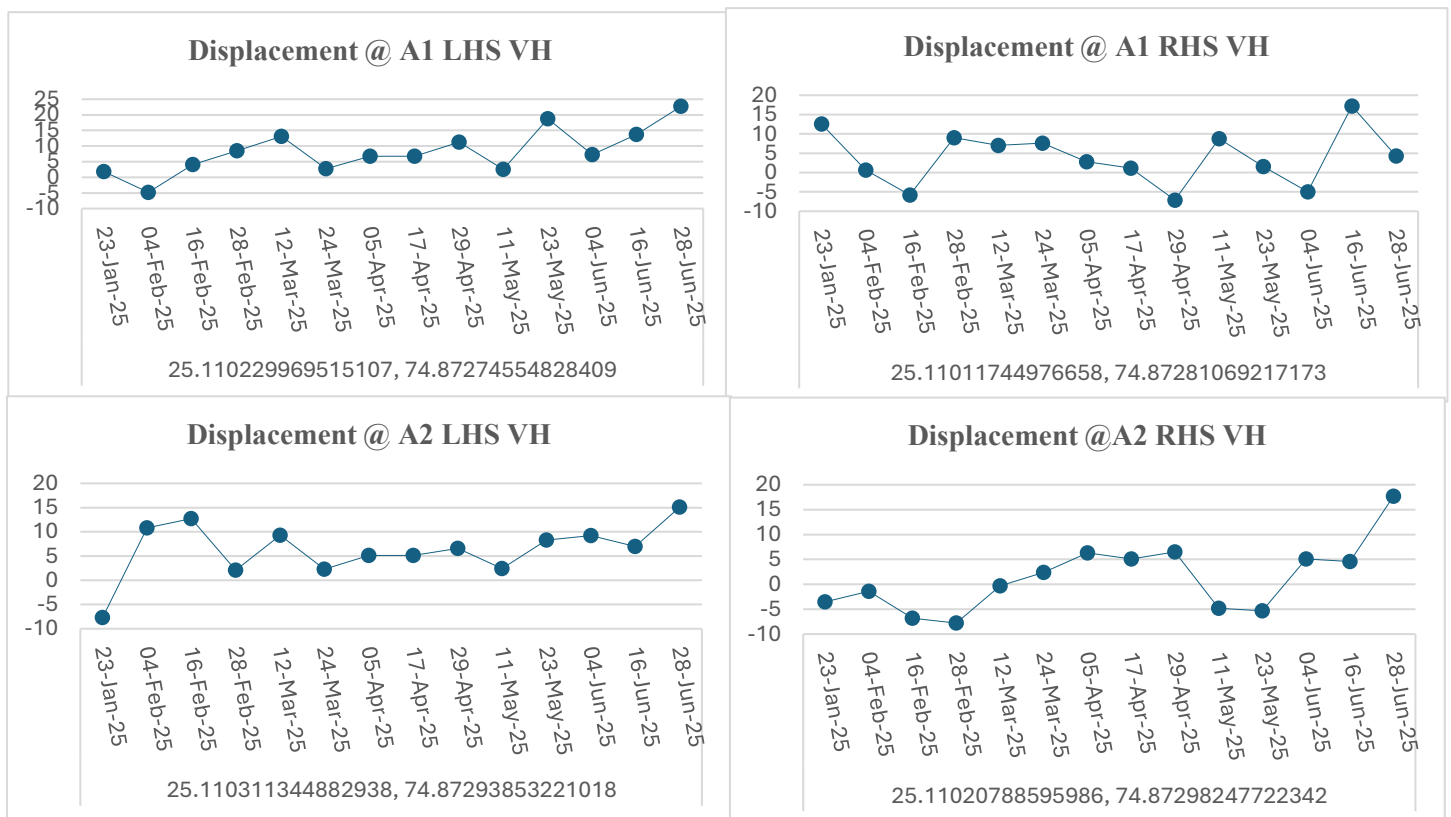


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

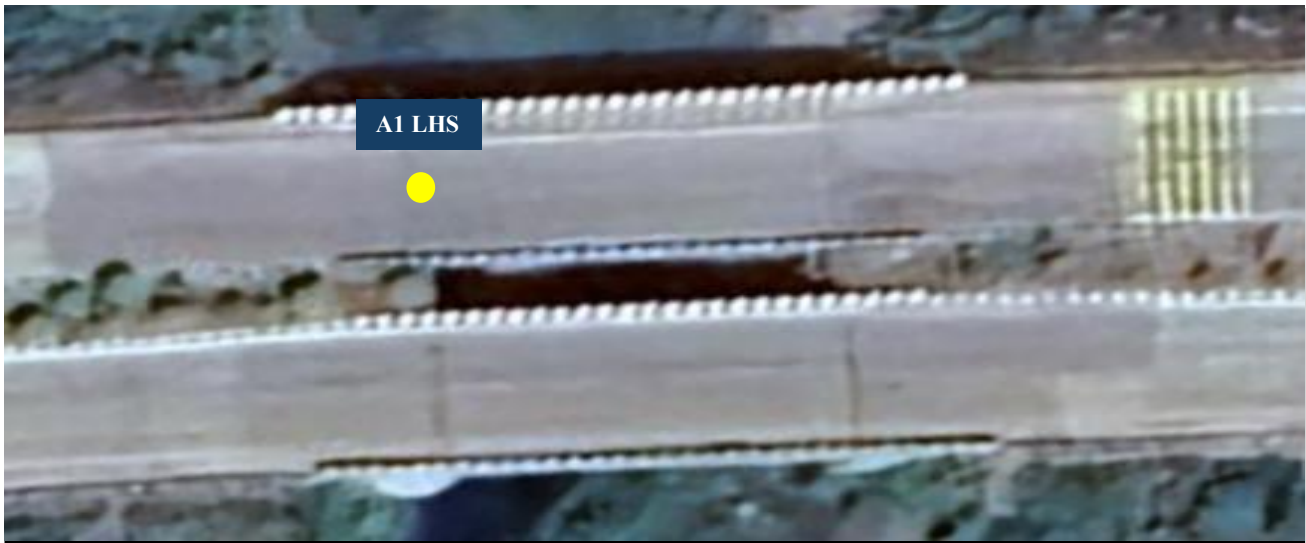
3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



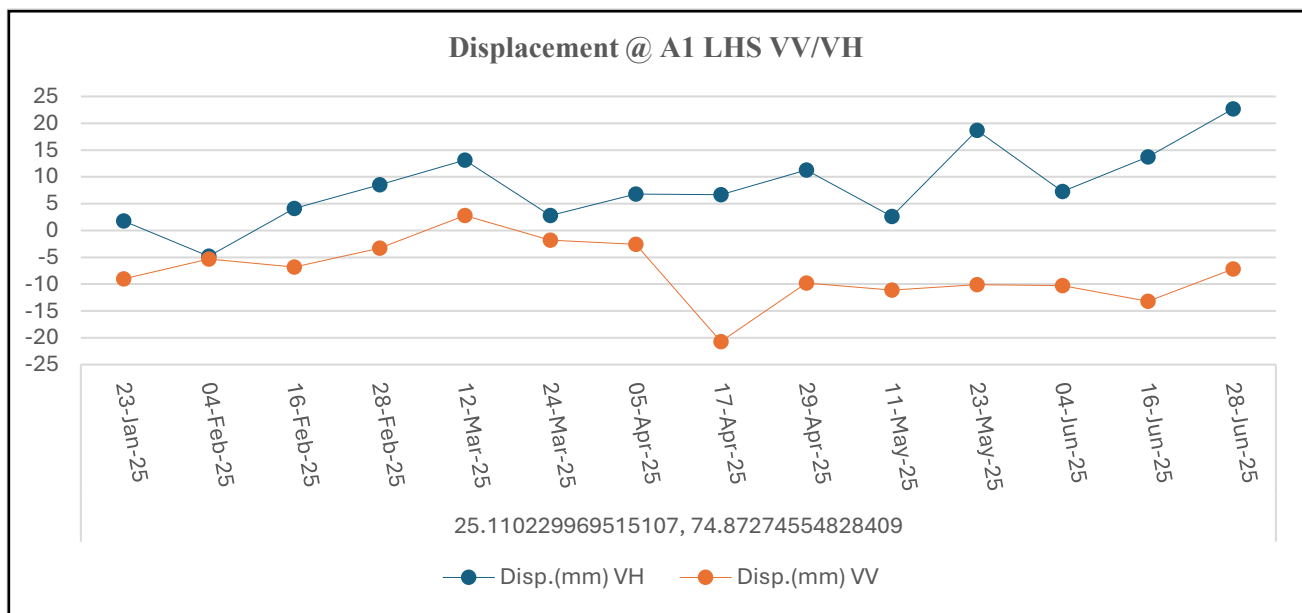
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.110229969515107, 74.87274554828409)



Key Observations:

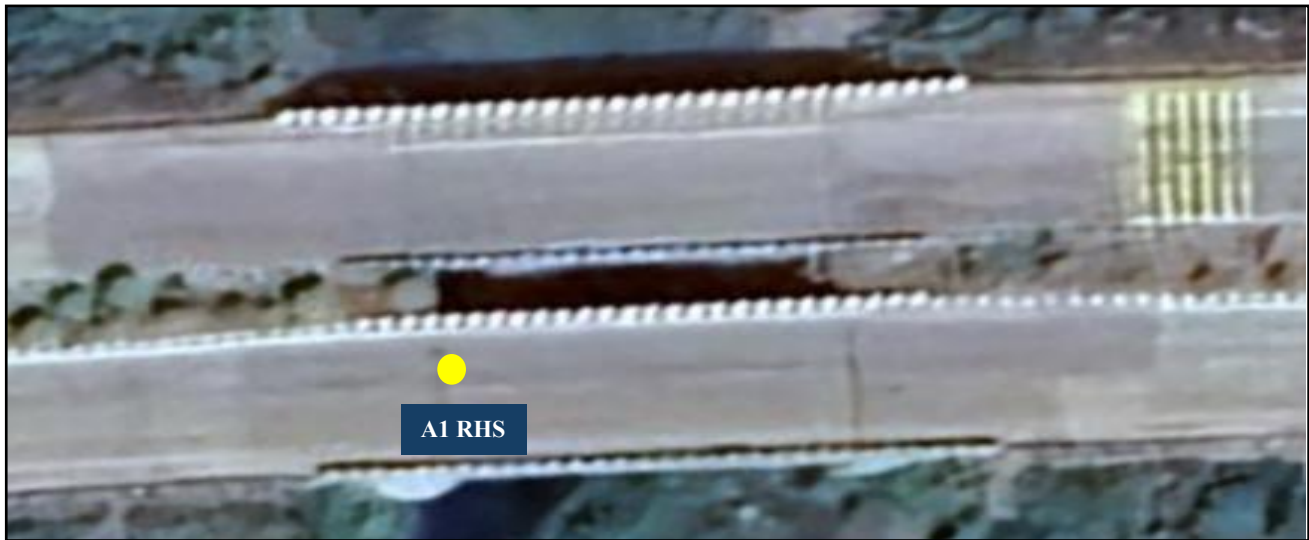
Displacement Magnitude Patterns:

- Vertical Displacement (VV)** remains consistently negative throughout the monitoring period, ranging approximately from -5 mm to -20 mm, indicating sustained downward settlement with one major dip in mid-April.
- Horizontal Displacement (VH)** shows a clear upward trend overall, starting near 0 mm in January 2025 and rising steadily toward 20–23 mm by late June 2025, reflecting progressively increasing lateral movement at A1 LHS.

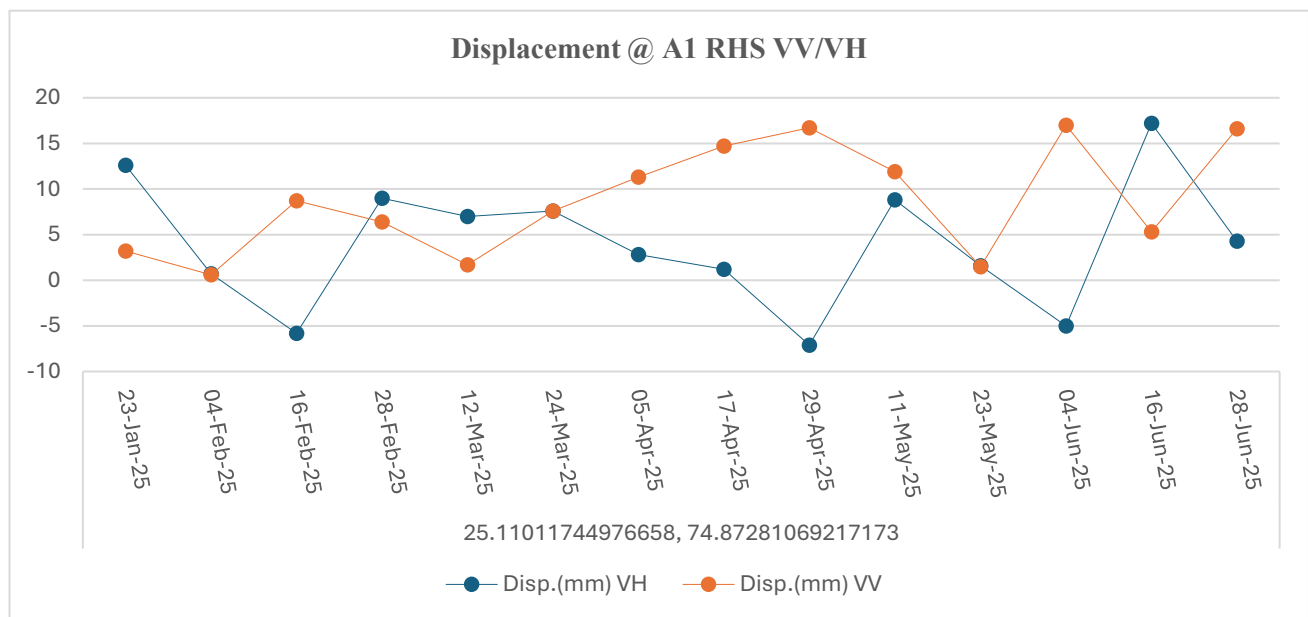
Critical Events & Anomalies:

- March 2025:** VH rises sharply to around 12–13 mm, marking the first significant increase in lateral displacement. VV remains moderately negative (around -5 mm), indicating stable vertical movement during this period.
- June 2025:** VH reaches its highest displacement level, marking the most significant lateral deformation observed, whereas VV trends upward slightly (less negative), indicating partial settlement recovery.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.11011744976658, 74.87281069217173)



Key Observations:

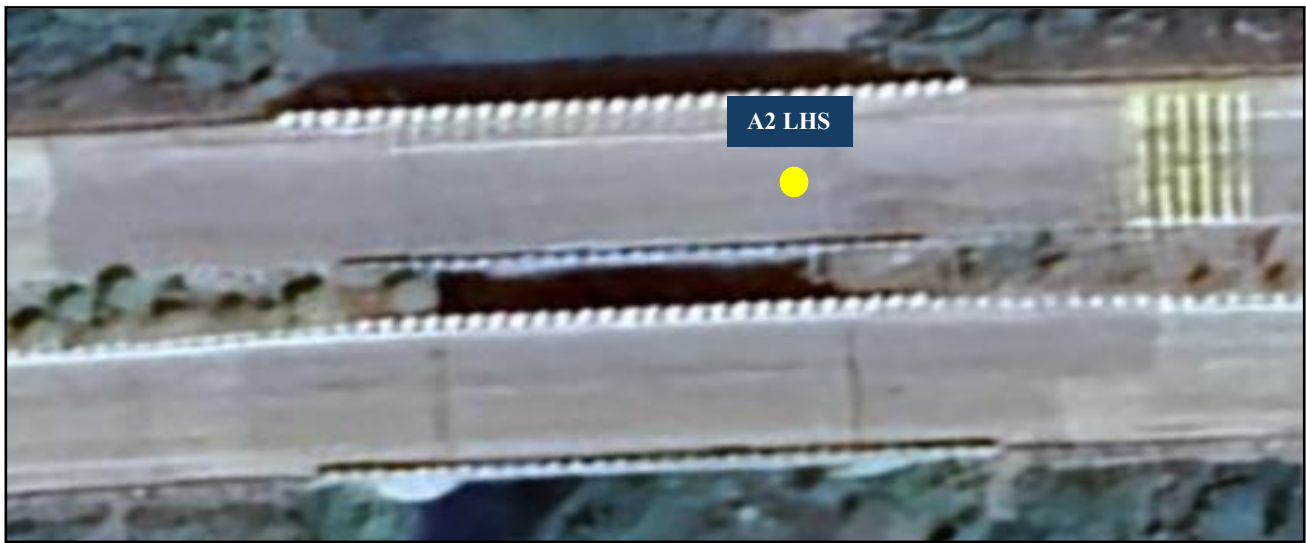
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** fluctuates considerably, ranging from approximately –7 mm to +17 mm, showing alternating periods of inward and outward lateral movement but with no strong long-term directional trend.
- **Vertical displacement (VV)** remains consistently positive, ranging approximately from +2 mm to +17 mm, indicating sustained upward vertical movement with several notable peaks throughout the monitoring period.

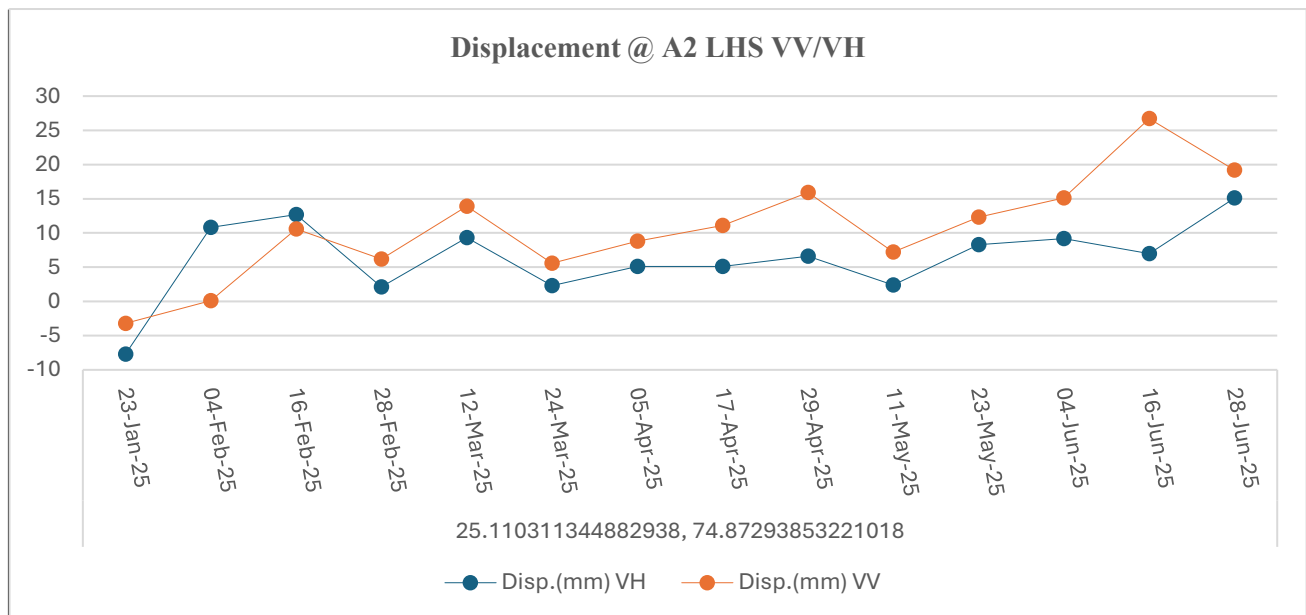
Critical Events & Anomalies:

- **February 2025:** VH drops sharply to around –7 mm, marking the most significant inward horizontal shift, while VV remains positive (+8–9 mm), indicating vertical uplift despite lateral reversal
- **May 2025:** VV decreases sharply to around +2 mm, while VH also dips to 0 mm, indicating a brief period of combined vertical and horizontal reduction.
- **June 2025:** VV surges again to approximately +16–17 mm, matching its highest values, whereas VH rises back to around +16–17 mm.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.110311344882938, 74.87293853221018)



Key Observations:

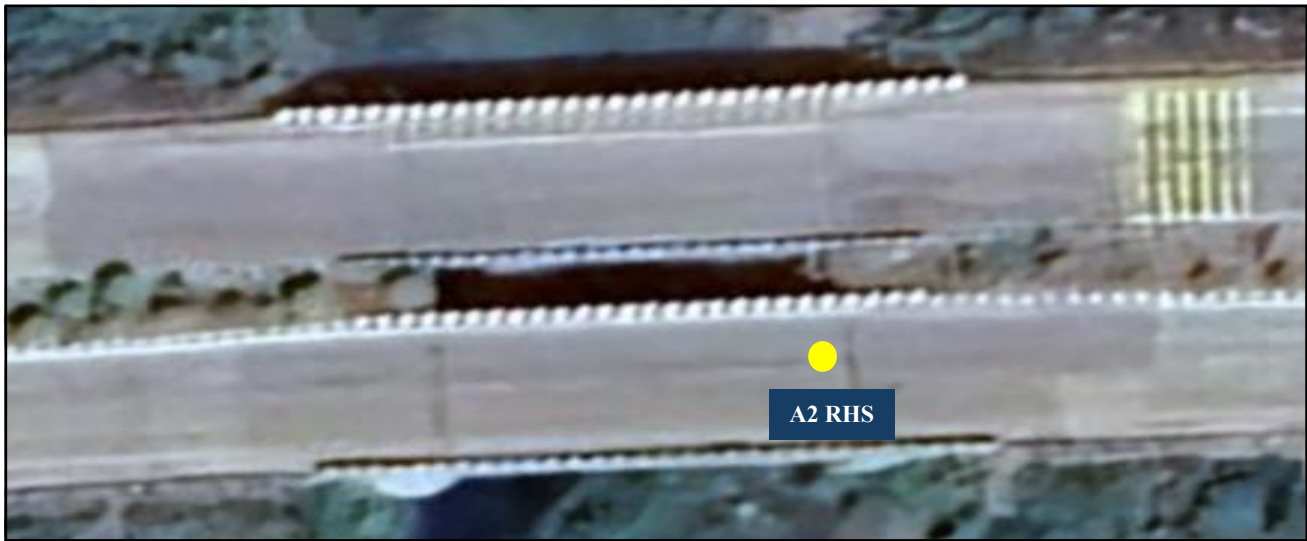
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** fluctuates but remains mostly positive, ranging from approximately 0 mm to +15 mm. VH shows gradual upward movement over time with alternating small dips and recoveries.
- **Vertical Displacement (VV)** shows a generally positive and increasing trend, ranging from around 0 mm in January 2025 to nearly +25 mm by late June 2025. This indicates progressive upward vertical movement with several intermediate peaks.

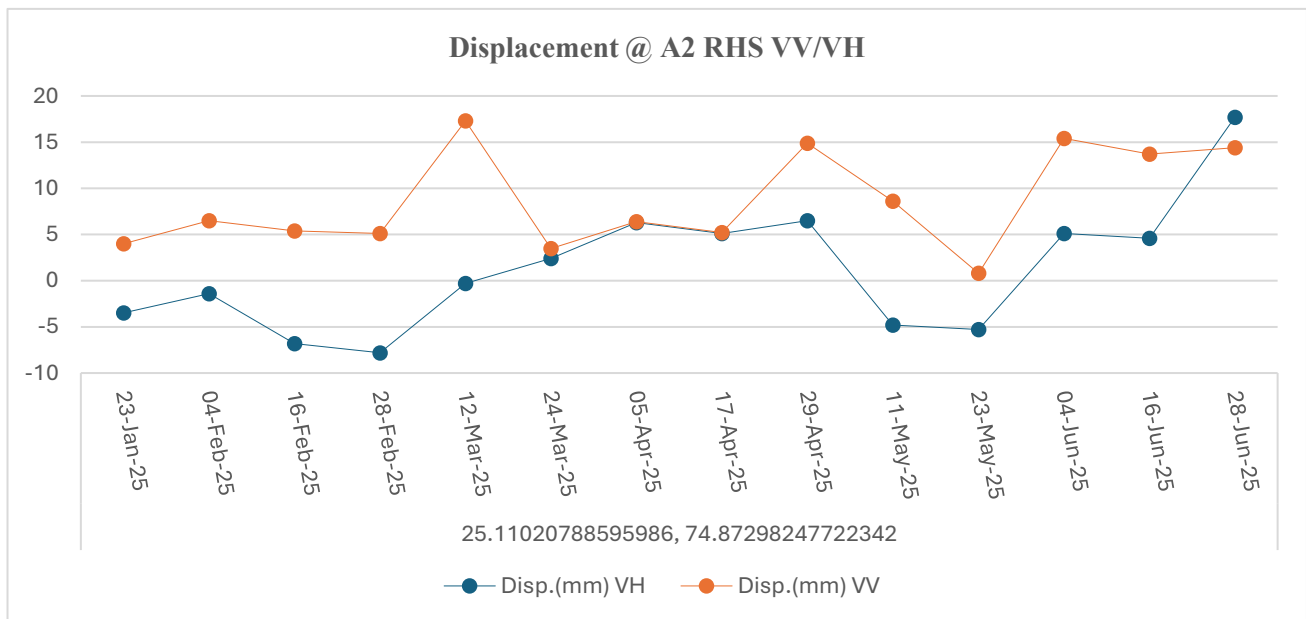
Critical Events & Anomalies:

- **February 2025:** Both VV and VH rise noticeably, with VV reaching approximately +12 mm and VH around +10 mm, marking an early dual-displacement increase on the A2 LHS.
- **June 2025:** VV reaches its highest displacement (+25 mm), while VH also climbs to around +14–15 mm, forming the strongest dual peak in both vertical and horizontal displacements for the A2 LHS location.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.11020788595986, 74.87298247722342)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** remains consistently positive, ranging approximately from +4 mm to +18 mm, indicating sustained upward vertical movement with multiple pronounced peaks (mid-March, late April, and mid-June).
- **Horizontal (VH) displacement** shows substantial fluctuation, ranging from approximately –8 mm to +18 mm, demonstrating alternating inward and outward lateral movements. While variable, VH shows an overall increasing trend toward the end of the monitoring period.

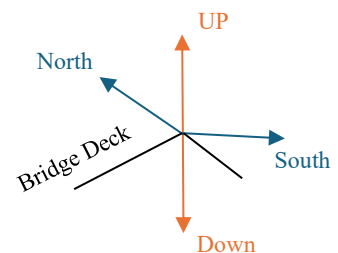
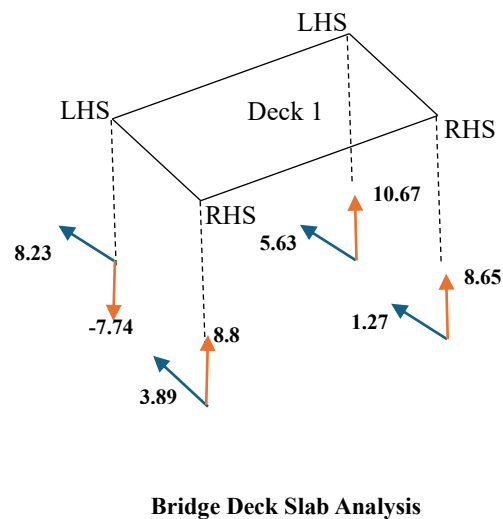
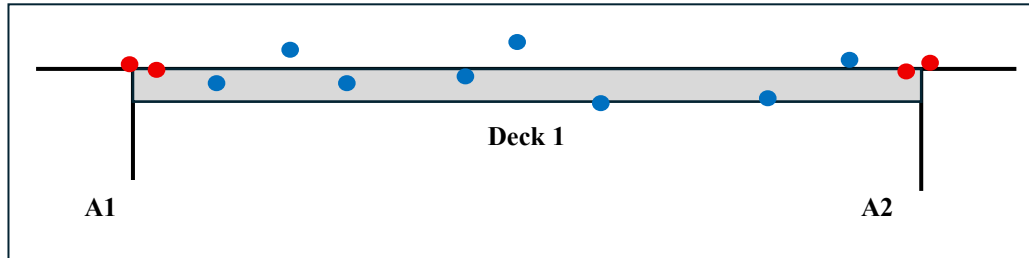
Critical Events & Anomalies:

- **February 2025:** VH dips sharply to around –8 mm, marking the strongest inward lateral shift during the period, while VV remains steady at around +6 mm, indicating vertical uplift despite horizontal reversal.
- **June 2025:** VV and VH both rise to their highest values (VV +16–17 mm, VH +18 mm), marking the strongest dual-displacement peak at A2 RHS.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The deck slab shows mixed upward and downward vertical movements, ranging from -7.74 mm downward to +10.67 mm upward, indicating non-uniform vertical response across both LHS and RHS locations.

Horizontal Displacement Behaviour:

- Horizontal movements range from 3.89 mm to 8.23 mm, with a clear lateral directionality across the slab.

CONCLUSION:

- The combined pattern of upward movement on the RHS, mixed uplift/settlement on the LHS, and horizontally divergent NW/NE directional shifts indicates torsional deformation of Deck 1 rather than uniform translation. Continued monitoring and structural evaluation are recommended to determine whether this torsional behaviour is progressive, stabilizing, or linked to bearing or support conditions.